ANTON BREDENBECK

Ph.D. Researcher

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EXPERIENCE

Ph.D. Researcher **TU Delft**

June 2022 – now Delft, Netherlands

- Supervised by Salua Hamaza and Cosimo Della Santina. I am conducting research on how drones can better exploit contact in guidance, navigation and control.
- Visiting researcher @UC Berkeley working with Mark W. Mueller on a tactile approach to high-speed collision recovery for UAVs.
- Actively involved in teaching through MSc student supervision, lectures, exercise development, and outreach activities.
- Member of the PhD council at the faculty of Aerospace Engineering supporting and representing the PhD community.

Research & Teaching Assistant **Julius-Maximilians University**

June 2020 – August 2021 Würzburg, Germany

- Worked on project DAEDALUS on hardware-and software of the robotic prototype for 3D point-cloud mapping of lunar caves.
- TA for "*Advanced Sensory Systems and Sensor Data Processing*".

Research Assistant **DFKI**

October 2020 – December 2020 Bremen, Germany

Worked on project Fast & Slow on software development for a ROS 2 interface to enable reinforcement learning of TurtleBots.

Working Student **Zentrum für Telematik e.V.**

October 2019 – December 2019 Würzburg, Germany

Validation, design and manufacturing of electronics for CubeSats

Research Assistant **Finnish Geospatial Research Institute**

March 2019 – August 2019 Helsinki, Finland

Development of a GNSS-Interference detection-and surveillance service (GNSS-Finland) and a "Dual Frequency Measurement Correction" System for Android Devices.

STRENGTHS

Hard-working | Resilient | Thorough | Structured

Robotics | Control | Estimation | Tactile Sensing | Design

C/C++ | Python | ROS 2 | ROS | MATLAB | Julia

EDUCATION

Ph.D. in Robotics

Biomorphic Intelligence Lab, TU Delft

June 2022 – Present Delft, Netherlands

I work on "*Tactile-driven Control for Aerial Robots*". Thereby I want to enable **aerial robots** to robustly and safely **interact** with the world around them, while exploiting contact as a source of information.

M.Sc. in Satellite Technology

Julius-Maximilians University

2019 – 2022 Würzburg, Germany

Graduated with honors (grade *1,0*).

- Six months specialization in Robotics & Control @CTU in Prague
- Master thesis @ESA/ESTEC working on trajectory optimization and control for a free-floating platform. The results were published at IROS 2022 in Kyoto.

B.Sc. in Aerospace Computer Science

Julius-Maximilians University

2016 – 2019 Würzburg, Germany

Six month internship at the Finnish Geospatial Research Institute in Helsinki, Finland.

REFEREES

Prof. Dr. Salua Hamaza
Biomorphic Intelligence Lab, Associate Professor, TU Delft
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Prof. Dr. Mark W. Mueller
HiPeR Lab, Associate Professor, UC Berkeley
mwm@berkeley.edu

Prof. Dr. Cosimo Della Santina
Cognitive Robotics, Associate Professor, TU Delft
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PUBLICATIONS

Journal Articles

- A. **Bredenbeck**, A. Jadoenathmisier, and S. Hamaza, "Aerial tactile perching via an anthropomorphic hand with embodied soft tactile receptors," *npj Robotics*, vol. 4, no. 1, p. 35, 2026. DOI: 10.1038/s44182-026-00109-9.
- D. Chaikalis, A. **Bredenbeck**, M. Brummelhuis, and S. Hamaza, "Sequential multi-tasking aerial manipulation with octopus-inspired tactile tentacles," *[under review] Nature Communications*, 2026.
- D. van Loon, A. **Bredenbeck**, L. Puck, M. Azkarate, and S. Hamaza, "Tether-inertial localization for planetary drones," *IEEE Robotics and Automation Letters*, vol. 11, 10 2026. DOI: 10.1109/LRA.2026.3723306.
- A. **Bredenbeck**, C. D. Santina, and S. Hamaza, "Embodying compliant touch on drones for aerial tactile navigation," *IEEE Robotics and Automation Letters*, vol. 10, no. 2, pp. 1209–1216, 2025. DOI: 10.1109/LRA.2024.3519888.
- A. **Bredenbeck**, T. Yang, S. Hamaza, and M. W. Mueller, "A tactile feedback approach to path recovery after high-speed impacts for collision-resilient drones," *Drones*, vol. 9, no. 11, 2025. DOI: 10.3390/drones9110758.
- F. Arzberger, J. Zevering, A. **Bredenbeck**, D. Borrmann, and A. Nüchter, "Unconventional trajectories for mobile 3d scanning and mapping," *Autonomous Mobile Mapping Robots*, 2022.
- F. Arzberger, A. **Bredenbeck**, J. Zevering, D. Borrmann, and A. Nüchter, "Towards spherical robots for mobile mapping in human made environments," *ISPRS Open Journal of Photogrammetry and Remote Sensing*, p. 100 004, 2021.

Conference Proceedings

- M. Schuster*, A. **Bredenbeck***, M. Beitelschmidt, and S. Hamaza, "Tactile odometry in aerial physical interaction," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2024)*, 2024.
- F. Arzberger, J. Zevering, A. **Bredenbeck**, D. Borrmann, and A. Nüchter, "Mobile 3d scanning and mapping for freely rotating and vertically descended lidar," in *2022 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*, IEEE, 2022, pp. 122–129.
- A. **Bredenbeck**, C. Della Santina, and S. Hamaza, "End-effector contact force estimation for aerial manipulators," in *IROS 2022 Workshop on Mobile Manipulation and Embodied Intelligence (MOMA): Challenges and Opportunities*, 2022.
- A. **Bredenbeck**, S. Vyas, W. Suter, *et al.*, "Finding and following optimal trajectories for an overactuated floating robotic platform," in *16th Symposium on Advanced Space Technologies in Robotics and Automation (ASTRA)*, 2022.
- A. **Bredenbeck**, S. Vyas, M. Zwick, D. Borrmann, M. Olivares-Mendez, and A. Nüchter, "Trajectory optimization and following for a three degrees of freedom overactuated floating platform," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2022, pp. 4084–4091.
- J. Zevering, D. Borrmann, A. **Bredenbeck**, and A. Nüchter, "The concept of rod-driven locomotion for spherical lunar exploration robots," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2022, pp. 5656–5663.
- D. Borrmann, A. Nüchter, A. **Bredenbeck**, *et al.*, "Lunar caves exploration with the daedalus spherical robot," in *52nd Lunar and Planetary Science Conference*, 2021, p. 2073.
- R. Pozzobon, A. Rossi, S. Ferrari, *et al.*, "Marius hills skylight hazard characterization as a possible landing site for lunar subsurface exploration," in *52nd Lunar and Planetary Science Conference*, 2021, p. 1886.
- A. P. Rossi, F. Maurelli, V. Unnithan, *et al.*, "Daedalus-descent and exploration in deep autonomy of lava underground structures: Open space innovation platform (osip) lunar caves-system study," 2021.
- J. Zevering, A. **Bredenbeck**, F. Arzberger, D. Borrmann, and A. Nüchter, "Imu-based pose-estimation for spherical robots with limited resources," in *2021 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, IEEE, 2021, pp. 1–8.
- S. Nikolskiy, A. **Bredenbeck**, T. Rikkinen, *et al.*, "Gnss signal quality monitoring based on a reference station network," in *2020 European Navigation Conference (ENC)*, IEEE, 2020, pp. 1–10.
- J. Zevering, A. **Bredenbeck**, F. Arzberger, D. Borrmann, and A. Nüchter, "Luna - a laser-mapping unidirectional navigation actuator," in *International Symposium on Experimental Robotics*, Springer, Cham, 2020, pp. 85–94.

GRANTS

- *DIGIT CFP*, Approximately USD 1,500, awarded in the form of four GelSight DIGIT sensors for research on tactile control of aerial robots, Meta AI, 2024.
- *North America Travel Grant*, EUR 4,500 awarded to selected TU Delft AI Labs PhD candidates to support research visits in North America. Used for a research visit to UC Berkeley to work with Mark W. Mueller, AI Lab Initiative, TU Delft, 2024.